

SHUBHAM GURAV

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Professional Summary

Generative AI Engineer with 1+ years experience in building and deploying LLM-powered applications. Skilled in prompt-tuning, RAG architectures, and real-time data retrieval, seeking to drive innovative AI solutions in a forward-thinking organization.

Professional Experience

VVT Solutions

Generative AI Developer

2023 – Present

Chennai, India

Key Projects: NEET Predictor, AI-Tutor , Question Bank System

- Successfully launched generative AI applications, such as intelligent **Chatbots**, to revolutionize the EdTech sector with projects like NEET Predictor and AI-Tutor.
- Develop and **Automated** MCQ creation system achieved a **90%** reduction in costs.
- Fine-tuned LLMs (**Llama3, Mistral**) utilized **quantization** and **LoRA** to reduce latency by 40%, improve performance by 30%, and accelerated fine-tuning by 2x using **unsloth**.
- Developed **Multi-agent systems, RAG pipelines** and implemented advanced **Prompt-engineering** techniques for educational applications, delivering personalized and interactive learning experiences.
- Enhanced backend integration with vector databases,MySQL and optimized data retrieval using cutting-edge frameworks like **LangChain** and **LlamaIndex**.
- Designed and deployed **Natural Language Processing (NLP)** solutions to streamline data analysis and enhance AI-driven functionalities.
- Wrote scalable, production-ready code for AI models and applications, with extensive use of **Docker** and **Deployment scripting** for efficient **DevOps** workflows.

SKILLS

Programming Languages: Python,R,SQL

Databases: MySQL,Astra DB , Pinecone

API Development : FastAPI ,Flask

Frameworks : Hugging Face,Langchain,LlamaIndex ,TensorFlow,Pytorch

AI/ML Expertise : Machine learning, Deep learning, Natural language processing (NLP), LLM fine-tuning, Quantization, PEFT, Prompt Engineering, RAG architectures,Mutli-Agent frameworks

MLOps: Git, Docker, CI/CD Pipelines, Model Deployment & Monitoring, AWS Sagemaker Pipelines, MLflow

Cloud Services : AWS (Sagemaker, EC2,ECR,S3)

Education

Visvesvaraya Technological University

Bachelor of Engineering

Aug 2018 – Sep 2022

Belgaum, karnataka

PROJECTS:

1.NEET Predictor

Overview: Developed an intelligent NEET predictor bot that **retrieves** data using **SQL** and **Astra DB**, providing personalized college suggestions and details.Integrated advanced retrieval systems using **Langchain** to store and retrieve college information in a **vector database**.

Technologies: **GPT 3.5,Assistant API, Langchain, MySQL, Astra DB,RAG**

Outcome: Delivered personalized,data-driven college recommendations.Increased accuracy by **30%** through **advanced vector search**. Reduced manual search time by **80%**

2.AI-Tutor

Overview : Developed a learning platform which engages students by delivering content, clarifying doubts, generating questions for evaluation and analyzing strengths and weaknesses and provides personalized study plans.

Technologies: **LLMs(GPT-4o, Gemini flash 1.5),Langchain,Llama index,Astra DB,Prompt Engineering**

Outcome: Delivered a highly interactive and **personalized** learning experience and adaptive study plans for each student.

3.Question Bank System

Overview: Developed a question bank system application that utilizes a **Chain of Thought (COT) prompt tuning,multi-agent framework** to generate, validate, and store high-quality MCQs in a **MySQL** database.

Technologies: **LLMs(Gemini, Claude), LangChain,Prompt Engineering ,MySQL**

Outcome: The automated system reduced the need of faculty **cutting costs by 90%**

COURSE AND CERTIFICATIONS

1. Data Science Certification

2.Tableau Desktop professional Certification